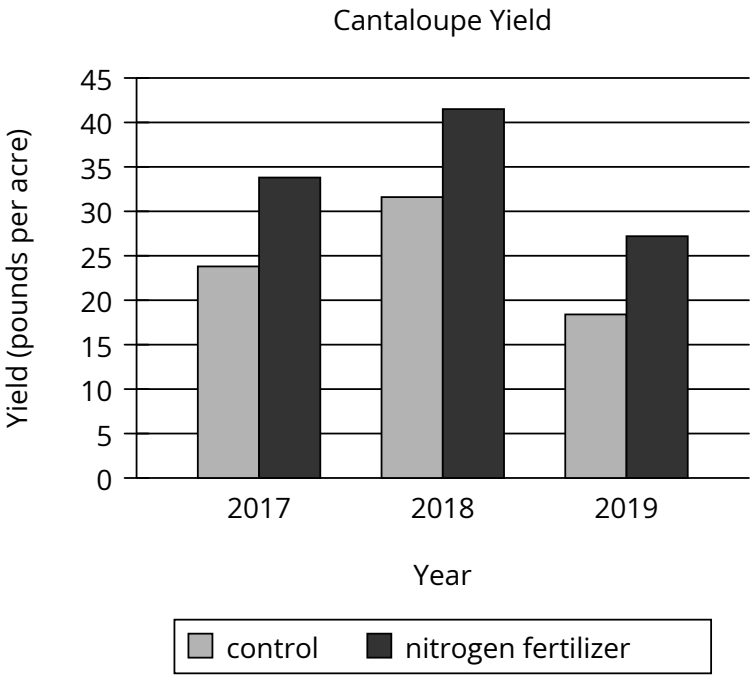


Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Medium

Question



To test the effects of a nitrogen fertilizer on cantaloupe production, researchers grew cantaloupe plants and harvested their fruit over three years. In each year, half the plants were grown using a nitrogen fertilizer, and the other half were grown using a control fertilizer that contained no nitrogen. The researchers concluded that the nitrogen fertilizer increases cantaloupe yield.

Which choice best describes data in the graph that support the researchers’ conclusion?

- Answer**
- A. In every year of the experiment, plants treated with the nitrogen fertilizer had a yield of at least 30 pounds per acre.
 - B. In every year of the experiment, plants treated with the nitrogen fertilizer had a greater yield than did plants treated with the control fertilizer.
 - C. The 2018 yield for plants treated with the control fertilizer was greater than was the 2019 yield for plants treated with the nitrogen fertilizer.
 - D. The yield for plants treated with the nitrogen fertilizer increased from 2017 to 2018.

Correct Answer: B

Rationale

Choice B is the best answer because it describes data from the graph that support the researchers’ conclusion that using nitrogen fertilizer increases cantaloupe production. The bar graph shows the cantaloupe yield for plants grown with nitrogen fertilizer and for those grown with a control fertilizer (without nitrogen) for three consecutive years (2017–2019). For each year in the graph, the yield for the nitrogen group is greater than the yield for the control group. In pounds per acre, the value in 2017 for the nitrogen-group yield is greater than 30 while the control-group yield is less than 25; in 2018 the nitrogen-group yield is greater than 40 while the control-group yield is less than 35; and in 2019 the nitrogen-group yield is greater than 25 while the control-group yield is less than 20. Thus, these data from the graph strongly support the conclusion that using nitrogen fertilizer increases cantaloupe yield.

Choice A is incorrect because the bar graph shows that in 2019 the nitrogen group had a yield below 30 pounds per acre. Choice C is incorrect. Although this choice accurately describes data in the bar graph—in 2018, the control-group yield is greater than 30 pounds per acre, and in 2019 the nitrogen-group yield is less than 30 pounds per acre—a claim that the control-group yield exceeds that of the nitrogen group strongly conflicts with the researchers' conclusion that nitrogen fertilizer produces larger yields. Choice D is incorrect. Although it is true that the bar graph shows a higher yield for the nitrogen group in 2018 than in 2017, without the control to compare against, it is impossible to know whether the increase is due to the fertilizer and not, for example, more favorable weather in 2018 than in 2017.